Vector-space containers (VCont)

A self-contained treatment: the biproduct collapse, its exact finite-field defect, algebraic structure, and the admissibility frontier

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Reading guide and honesty note. Every theorem below is tagged in prose with its epistemic grade: **proved** (a complete pen-and-paper proof exists in the cited internal file), **lean-verified** (machine-checked, `sorry`-free, in Lean 4), or **OPEN** (a precisely stated conjecture that is *not* settled). The headline result of §2 is proved and lean-verified. The frontier question of §4 ("Question (Q)") is **OPEN** and is conjectured independent of ZFC; it is presented as such and nowhere treated as settled. This note has not yet had a neighbour review.

Fix a field k. A *container* in a category $\mathcal{C}$ is a pair (shapes, positions): a set S together with a position object $P_s \in \mathcal{C}$ for each $s \in S$, extending to an endofunctor $[\![S, P]\!]$ of $\mathcal{C}$. Over $\mathcal{C} = \mathbf{Set}$ this is the classical theory of Abbott, Altenkirch and Ghani: containers are precisely the polynomial functors, and the extension functor is a full and faithful embedding. This note asks what survives when the positions are *vector spaces* — the base $\mathcal{C} = \mathbf{Vec}_{\mathrm{fd}}$ of finite-dimensional k-vector spaces — and reports the answers the programme has established. The one-line summary: *the coproduct/product distinction that makes* **Set** *containers rigid becomes a biproduct over* **Vec**, *and much of what the* **Set** *theory reads off "for free" collapses.* Over a finite field $\mathbb{F}_q$ that collapse is not merely qualitative: §3 computes its exact size $\delta = \prod_s a_s - \prod_s(a_s - (|T| - 1))$ as a closed integer point-count.

1 Definition of VCont and its extension functor

Write $\mathbf{Vec} = \mathbf{Vec}_k$ for all k-vector spaces, $\mathbf{Vec}_{\mathrm{fd}}$ for the finite-dimensional ones, and $\mathbf{Vec}(P, W)$ for the hom vector space. All functors below are $\mathbf{Vec}$-functors: their action on homs $\mathbf{Vec}(V, W) \to \mathbf{Vec}(FV, FW)$ is k-linear.

Definition 1 (linear / vector-space container)**.** A *vector-space container* (object of $\mathbf{VCont} :=$ $\mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})$, or of $\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ when finiteness is imposed) is a pair $(S, (P_s)_{s \in S})$ with S a set (the *shapes*) and each $P_s \in \mathbf{Vec}$ a *position space*. A morphism $(S, (P_s)) \to (T, (Q_t))$ is a pair $(f, (\varphi_s))$ with $f\colon S \to T$ a function and, for each $s \in S$, a linear map $\varphi_s\colon Q_{f(s)} \to P_s$ — *covariant on shapes, contravariant on positions*, exactly as for **Set** containers. Write $n_s := \dim P_s$.

Definition 2 (extension)**.** The *extension* of (S, P) is the functor $[\![S, P]\!]\colon \mathbf{Vec} \to \mathbf{Vec}$,

$$[\![S, P]\!]\, W \;:=\; \bigoplus_{s \in S} \mathbf{Vec}(P_s,\, W).$$

Choosing a basis of P_s gives $\mathbf{Vec}(P_s, W) \cong W^{n_s}$ naturally in W, so $[\![S, P]\!]\, W \cong \bigoplus_{s \in S} W^{\oplus n_s}$, a "direct sum of powers". On a morphism $(f, (\varphi_s))$ the natural transformation is $(g_s)_s \mapsto \big(g_{f(s)} \circ \varphi_s\big)_s$.

Writing $h_P := \mathbf{Vec}(P, -)$ for the corepresentable at P, we have $[\![S, P]\!] = \bigoplus_{s \in S} h_{P_s}$ and $h_k = \mathbf{Vec}(k, -) \cong \mathrm{Id}$.

Two features of **Vec** shape everything that follows.

Coproduct = product = biproduct. A finite coproduct in **Vec** is a biproduct, hence *equals* the product. Thus although the outer sum $\bigoplus_{s \in S}$ in Definition 2 is written as a coproduct (a "Σ over shapes" of the hom-spaces $\mathbf{Vec}(P_s, X)$), for finite S its underlying space is equally the product $\prod_{s \in S} \mathbf{Vec}(P_s, X)$ — a "Π over shapes". This $\Sigma \rightsquigarrow \Pi$ coincidence is the *biproduct collapse*, and it is the single change from the set-level extension $[\![S, P]\!]X = \coprod_{s \in S} X^{P_s}$ (a genuine disjoint union). It is made fully explicit in the Lean formalisation of §6, where the linear extension is literally `LExt` $(S, \mathrm{Pos})\, X = (s : S) \to (\mathrm{Pos}\, s \to X)$.

Finite positions distribute over $\oplus$. If $\dim P < \infty$ then $\mathbf{Vec}(P, \bigoplus_t V_t) \cong \bigoplus_t \mathbf{Vec}(P, V_t)$ naturally; equivalently, h_P preserves coproducts iff P is finite-dimensional (a finite-support argument on a basis of P). This is why we usually restrict positions to $\mathbf{Vec}_{\mathrm{fd}}$; the infinite-dimensional case is the source of the frontier in §5.

Example 1. Take $S = \{a, b\}$ with $P_a = k^2$, $P_b = k$. Then $[\![S, P]\!]\, W = \mathbf{Vec}(k^2, W) \oplus \mathbf{Vec}(k, W) \cong W^2 \oplus W \cong W^3$, so $[\![S, P]\!] \cong \mathrm{Id}^3$. Numerically $\dim[\![S, P]\!](k^d) = 3d$ for every d (verified computationally, $d = 0, \ldots, 4$). The two integers that produced this functor — the shape S and the partition $3 = 2 + 1$ — have already vanished from the answer: only the total $N = \sum_s n_s = 3$ remains. That disappearance is the subject of §2.

2 The collapse: $[\![-]\!]$ is neither full nor faithful over Vec, and not injective on objects

Over **Set** the extension functor $[\![-]\!] \colon \mathrm{Fam}(\mathbf{Set}^{\mathrm{op}}) \to [\mathbf{Set}, \mathbf{Set}]$ is **full and faithful** (Abbott–Altenkirch–Ghani): the container data (S, P) can be recovered from the functor, with $S = [\![S, P]\!](1)$ the value at a one-element set. Over **Vec** this recovery fails at its root.

Remark 1 (the terminal slogan dies). The terminal object of **Vec** is the zero object 0, so $[\![S, P]\!](0) = \bigoplus_s \mathbf{Vec}(P_s, 0) = 0$: the shapes are invisible at the terminal object, and the **Set** slogan "$F(1) = S$" has no analogue.

2.1 The finite collapse (headline; proved and lean-verified)

Theorem 1 (finite collapse; **proved**). *Let S be finite and every $n_s = \dim P_s$ finite, and put $N := \sum_{s \in S} n_s = \dim\big(\bigoplus_s P_s\big)$. Then there is a natural isomorphism $[\![S, P]\!] \cong \mathrm{Id}^N$. Consequently:*

1. *$[\![S, P]\!] \cong [\![S', P']\!]$ as functors **iff** $N = N'$; a finite vector-space container is classified up to isomorphism of its extension by the single number N.*

2. *Since $\mathrm{End}(\mathrm{Id}) = k$ is local and $\mathrm{End}(\mathrm{Id}^N) = M_N(k)$ is semiperfect, Krull–Schmidt–Azumaya makes the decomposition into indecomposables unique; the recovered invariant is exactly the multiplicity N. The shape set S and the partition $N = \sum_s n_s$ are* not *recoverable.*

This is proved in `proofs/2026-08-18-linear-containers-vec.md` (Theorem 2.1; registry `linear-containers-vec.json`, node `part1-finite-collapse`, grade **proved**), with the co-Yoneda lemma $\mathrm{Nat}(h_P, G) \cong G(P)$ and the additivity of $P \mapsto h_P$ as the engine.

Theorem 2 (the explicit witness; **proved**, **lean-verified**). *The two containers*

$$p = (\{*\},\, k^2) \qquad \text{and} \qquad p' = (\{1,2\},\, (k,k))$$

both have extension Id^2*, i.e. both present the functor* $X \mapsto X \oplus X$*, yet they are **not isomorphic as containers**: their shape sets have different cardinalities* ($1 \neq 2$)*. Hence* $[\![-]\!]$ *is* not injective on objects.

Contrast **Set**: there the analogous containers $(\{*\}, 2)$ and $(\{1,2\},(1,1))$ have *distinct* extensions X^2 and $2 \cdot X$, separated already at $X = 1$. The collapse is precisely the failure of that separation. Theorem 2 is machine-checked; see §6.

2.2 Why: the hom formula and the failure of extensivity

The mechanism is visible one categorical level up, on hom-sets. Both computations below are proved in `2026-08-18-linear-containers-vec.md` (Theorems 3.1–3.3; registry nodes `part2-hom-formula`, `part2-extensivity-crux`, both **proved**).

Proposition 1 (hom formula; **proved**). *For vector-space containers* (S,P)*,* (T,Q)*,*

$$\begin{aligned} \mathrm{Nat}\big([\![S,P]\!],[\![T,Q]\!]\big) \;&\cong\; \prod_{s\in S}\,\bigoplus_{t\in T}\mathbf{Vec}(Q_t,P_s),\\ \mathrm{Fam}(\mathbf{Vec}^{\mathrm{op}})\big((S,P),(T,Q)\big) \;&\cong\; \prod_{s\in S}\,\coprod_{t\in T}\mathbf{Vec}(Q_t,P_s), \end{aligned}$$

where $\coprod$ *is the coproduct in* **Set** *(disjoint union): a container morphism assigns to each* s *a single choice* $t = f(s)$ *together with a linear map in* $\mathbf{Vec}(Q_{f(s)},P_s)$.

Theorem 3 (neither full nor faithful; **proved**). *The functor* $[\![-]\!]$ *induces on each hom the canonical comparison*

$$\prod_s \coprod_t \mathbf{Vec}(Q_t,P_s) \;\longrightarrow\; \prod_s \bigoplus_t \mathbf{Vec}(Q_t,P_s),$$

componentwise the map from the disjoint union $\coprod_t$ *into the biproduct* $\bigoplus_t$ *sending a choice* (t,φ) *to the element supported at* t*. Over* **Vec** *this map is* neither surjective nor injective*, so* $[\![-]\!]$ *is **neither full nor faithful**.*

1. Not full *as soon as some shape* s *admits two shapes* $t \neq t'$ *with* $\mathbf{Vec}(Q_t,P_s) \neq 0 \neq \mathbf{Vec}(Q_{t'},P_s)$*: any natural transformation whose* s*-component is a genuine linear combination* $\varphi_t + \varphi_{t'}$ *supported on two shapes is not in the image of a shape-function.*

2. Not faithful *as soon as* $S \neq \varnothing$ *and* $|T| \geq 2$*. Over a field the zero linear map lies in every* $\mathbf{Vec}(Q_t,P_s)$*, so the two* distinct *container morphisms* $(u,0)$ *and* $(u',0)$ *— with* $u \neq u' \colon S \to T$ *and all position maps zero — are collapsed to the* same *(zero) natural transformation. The mechanism is exactly the biproduct:* $\bigoplus_t$ *has a single shared zero across all summands, so it merges the* $|T|$ *distinct disjoint-union-tagged zeros* $(t,0)$ *into that one zero element; the comparison is injective only* away from the zero maps*. Hence* $[\![-]\!]$ *is **faithful iff*** $|T| \leq 1$ ***or*** $S = \varnothing$*. By contrast* **Set***'s coproduct is the disjoint union, which keeps a distinct tag on each zero, so the classical Abbott–Altenkirch–Ghani* $[\![-]\!]$ *over* **Set** *remains faithful.*

Remark 2 (provenance of the faithfulness correction). The failure of faithfulness was surfaced by R. Langer's finite-field analogue (q-container Cor. 5.3, Ex. 5.4); over $\mathbb{F}_q$ it is a finite point-count, made exact in §3. (Draft note.)

The point is a single symbol. Over **Set** the same computation gives $\mathrm{Nat}(\llbracket S,P\rrbracket,\llbracket T,Q\rrbracket)\cong\prod_s\bigoplus_t\mathbf{Set}(Q_t,P_s)$ where now $\bigoplus_t=\coprod_t$ is the coproduct *in* **Set**, i.e. the disjoint union: container-hom equals Nat, and $\llbracket -\rrbracket$ is full and faithful. **Set is extensive** — its coproduct is the disjoint union — and that is exactly what makes $\llbracket -\rrbracket$ full. **Vec is not extensive** — its coproduct is a biproduct — and the fullness gap

$$\coprod_t \subsetneq \bigoplus_t$$

is the precise, quantified manifestation of that failure. The object-level collapse (Theorem 1) and the morphism-level fullness gap (Theorem 3) are *one* phenomenon: the failure of extensivity under the base change $\mathbf{Set}\rightsquigarrow\mathbf{Vec}$.

Remark 3 (what survives; **proved**). The collapse used both finiteness of S and finite-dimensionality of positions; drop either and content returns. With infinite S and all $P_s=k$, $\llbracket S,P\rrbracket=\mathrm{Id}^{\oplus|S|}$ is a genuine coproduct, not a product, of copies of Id; and for a single infinite-dimensional P, $h_P=\mathbf{Vec}(P,-)$ is an infinite-product functor, not built from Id under $\oplus$, and P *is* recovered from the representable (Lemma 1.2/1.3 of the source file). The finite recovery is sharp: $\llbracket -\rrbracket$ determines the total space $\bigoplus_s P_s$ up to isomorphism, and nothing about its partition into shapes (Corollary 3.4, **proved**).

3 The finite-field instance: q-containers and the exact collapse defect

Everything in §2 is stated over an arbitrary field k and is *qualitative*: $\llbracket -\rrbracket$ either is or is not full, is or is not faithful. Over a *finite* field $k=\mathbb{F}_q$ these failures become finite point-counts, and the size of the collapse can be written in closed form. This is the setting of R. Langer's *q-container* note (personal communication, 2026-09-11): Definition 1 verbatim, with positions finite-dimensional $\mathbb{F}_q$-spaces. His §4 morphism count and §5 faithfulness corollary are the first two displays below; the exact *defect* (Theorem 4) is a joint refinement. This subsection is thus the arithmetic shadow of the whole of §2: the one structural symbol $\coprod_t\rightsquigarrow\bigoplus_t$ becomes the one integer $-(|T|-1)$.

Fix $\mathbb{F}_q$ and q-containers $C=(S,(P_s))$, $D=(T,(Q_t))$ with S,T finite; write $d_s=\dim P_s$, $e_t=\dim Q_t$, and $\mathbf{Cont}_q(C,D):=\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})\big((S,P),(T,Q)\big)$ for the (now finite) morphism set. The extension map on this hom, $\llbracket -\rrbracket\colon\mathbf{Cont}_q(C,D)\to\mathrm{Nat}(\llbracket C\rrbracket,\llbracket D\rrbracket)$, has finite source and target, and Proposition 1 specialises to two counts:

$$|\mathbf{Cont}_q(C,D)|=\prod_{s\in S}\Big(\sum_{t\in T}q^{\,e_td_s}\Big),\qquad |\,\mathrm{Nat}(\llbracket C\rrbracket,\llbracket D\rrbracket)|=q^{\,(\sum_s d_s)(\sum_t e_t)}.\tag{1}$$

The first is a product of independent per-shape choices: at each s pick a target $u(s)=t$ and a map in $\mathbf{Vec}(Q_t,P_s)$, of which there are $q^{e_td_s}$. The second is co-Yoneda: with $P:=\bigoplus_s P_s$ and $Q:=\bigoplus_t Q_t$ the extension is corepresentable, $\llbracket C\rrbracket=\mathbf{Vec}(P,-)$, so $\mathrm{Nat}(\llbracket C\rrbracket,\llbracket D\rrbracket)\cong\mathbf{Vec}(Q,P)$, of cardinality $q^{\dim Q\cdot\dim P}$.

Under that co-Yoneda bijection a morphism $(u,(f_s))$ becomes the block matrix $M\in\mathbf{Vec}(Q,P)$, rows indexed by s and columns by t, whose s-th row carries the single block f_s in column $u(s)$ and zero elsewhere. So as (u,f) ranges over $\mathbf{Cont}_q(C,D)$ the induced matrix is built *row by row, independently across rows*, and each row g_s ranges over the set of tuples supported in a single column:

$$A_s\;=\;\bigcup_{t\in T}W_t^{(s)},\qquad W_t^{(s)}:=\{\,g\in\textstyle\bigoplus_{t'}\mathbf{Vec}(Q_{t'},P_s):g_{t'}=0\ (t'\neq t)\,\}\;\cong\;\mathbf{Vec}(Q_t,P_s).$$

The image of $\llbracket - \rrbracket$ is therefore the product $\prod_s A_s$, and the only nontrivial count is $|A_s|$: a union of $|T|$ coordinate subspaces of a direct sum, which pairwise meet only in 0.

Lemma 1 (union of pairwise-trivially-meeting subspaces; **proved**). *If $W_1, \ldots, W_n$ ($n \geq 1$) are finite subspaces of an $\mathbb{F}_q$-space with $W_i \cap W_j = \{0\}$ for $i \neq j$, then $\left|\bigcup_i W_i\right| = \left(\sum_i |W_i|\right) - (n-1)$.*

Each W_i contains 0; for $i \neq j$ the only shared vector is 0, so the punctured sets $W_i \setminus \{0\}$ are pairwise disjoint and $\left|\bigcup_i W_i\right| = 1 + \sum_i (|W_i| - 1) = (\sum_i |W_i|) - (n-1)$. Applied to the $|T|$ coordinate subspaces $W_t^{(s)}$, with $|W_t^{(s)}| = q^{e_t d_s}$, this gives (for $|T| \geq 1$) $|A_s| = \left(\sum_t q^{e_t d_s}\right) - (|T| - 1)$.

Theorem 4 (collapse-defect formula; **proved**). *Write $a_s := \sum_{t \in T} q^{e_t d_s}$. For q-containers $C = (S, (P_s))$, $D = (T, (Q_t))$ with $|T| \geq 1$,*

$$\left|\,\mathrm{image}\,\llbracket - \rrbracket\,\right| = \prod_{s \in S} \big(a_s - (|T|-1)\big), \qquad \delta(C, D) := |\mathbf{Cont}_q(C, D)| - \left|\,\mathrm{image}\,\llbracket - \rrbracket\,\right| = \prod_s a_s - \prod_s \big(a_s - (|T|-1)\big).$$

Combine $|\mathrm{image}\,\llbracket - \rrbracket| = \prod_s |A_s|$ with Lemma 1 and subtract (1). Proved in `proofs/2026-09-11-collapse-defect-formula.md` (Theorem 5; registry `linear-containers-vec.json`, node `collapse-defect-formula`, grade **proved**), brute-force verified on 15 cases over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_4$ (dims ≤ 2, $|S|, |T| \leq 3$; `scratch/collapse_defect_verify.py`), all exact.

The formula makes the two qualitative statements of §2 into arithmetic:

Corollary 1 (faithful/full, quantified; **proved**). *For $|T| \geq 1$: $\llbracket - \rrbracket$ is faithful $\iff$ $\delta(C, D) = 0$ $\iff$ $|T| = 1$ or $S = \varnothing$ (the threshold of Theorem 3, now read off as "every factor $a_s - (|T| - 1)$ equals a_s"); and $\llbracket - \rrbracket$ is full $\iff$ every factor A_s fills its space $\iff$ for each shape s with $P_s \neq 0$ at most one target t has $Q_t \neq 0$.*

The single term $-(|T| - 1)$ is the whole story. It counts exactly the $|T|$ distinct disjoint-union tags $(t, 0)$ that the biproduct $\bigoplus_t$ merges into its one shared zero: the $|T|$ morphisms $(u, f{=}0)$ with different shape-functions $u \colon S \to T$ all induce the same zero row. Over **Set** the corresponding union is a genuine disjoint union $\coprod_t$, which keeps a distinct tag on every zero; the correction term vanishes, $\delta = 0$, and the Abbott–Altenkirch–Ghani representation is full and faithful. Thus the finite-field defect is the failure of extensivity (§2) rendered as an integer.

Remark 4 (the $q \to 1$ limit does *not* recover **Set**-faithfulness; **proved**). It is tempting to read δ through a "field with one element". Formally $a_s \to |T|$ and $a_s - (|T| - 1) \to 1$, so $|\mathrm{image}| \to 1$ while $|\mathbf{Cont}| \to |T|^{|S|}$: the $q \to 1$ degeneration is *maximal* collapse, $\delta \to |T|^{|S|} - 1 \neq 0$, not the faithful **Set** case. The reason is structural: the **Set** extension $V \mapsto \coprod_s \mathbf{Set}(P_s, V)$ is a *different functor* (a coproduct of representables), full and faithful by AAG; the linear extension replaces that **Set**-coproduct by the biproduct. The $q \to 1$ limit reflects only that every hom-space collapses over $\mathbb{F}_1$, not any set-level faithfulness. The two functors must not be conflated.

The witness that surfaced the whole faithfulness correction is the smallest nontrivial instance: $C = (\{*\}, k)$, $D = (\{1, 2\}, (k, k))$ over $\mathbb{F}_2$ gives $|\mathbf{Cont}| = 4$, $|\mathrm{image}| = 3$, $\delta = 1$ — two of the four morphisms (the two zero maps with $u = 1$ vs. $u = 2$) collapse to one natural transformation (Langer, Ex. 5.4). A sample of the verified table:

q	C	D	$\lvert\mathbf{Cont}\rvert$	$\lvert$image$\rvert$	$\lvert\,$Nat$\,\rvert$	δ
2	(1)	$(1,1)$	4	3	4	1
2	$(1,1)$	$(1,1)$	16	9	16	7
2	(1)	$(1,1,1)$	6	4	8	2
2	(2)	$(1,1)$	8	7	16	1
3	$(1,1)$	$(1,1)$	36	25	81	11
2	$(1,1)$	(2)	16	16	16	0 ($\lvert T\rvert$=1)

Here C, D are written as the tuples of position dimensions (d_s), (e_t). The $|T| = 1$ row realises Corollary 1's faithful case ($\delta = 0$); the $(2) \to (1,1)$ row is the Theorem 2 witness direction. The finite-dimensional theory over a finite field is thus not merely qualitatively but *numerically* pinned down.

4 Algebraic structure

Composition of vector-space containers is available on the finite-dimensional locus, and it too is shaped by the collapse.

Proposition 2 (composition; **proved**). *For containers with finite-dimensional positions,*

$$[\![S, P]\!] \circ [\![T, Q]\!] \;\cong\; [\![\, S \times T,\;\; (P_s \otimes Q_t)_{(s,t)} \,]\!],$$

so on $\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *the composition product is* $(S, P) \lhd (T, Q) = (S \times T, (P_s \otimes Q_t))$*, with unit* $I = (\{*\}, k)$ *(extension* Id*).*

This is *not* the **Set** container composition, whose shapes form the dependent sum $\coprod_{s \in S} T^{P_s}$: linearity flattens the dependency (a linear map $P_s \to \bigoplus_t V_t$ is a *sum of components*, not a *choice of branch*) into the plain product $S \times T$. Proved in `2026-08-18-linear-containers-vec.md`, Prop. 4.1 (registry node `part3-composition`, **proved**).

The comonoids for this product identify a clean algebraic species.

Theorem 5 ($\lhd$-comonoids are families of algebras; **proved**). *Let* $C = (S, P) \in \mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})$ *with* S *an arbitrary set and each* P_s *finite-dimensional. The* $\lhd$*-comonoid structures* (δ, ε) *on* C *in* $(\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}}), \lhd, I)$ *are in natural bijection with families* $(A_s)_{s \in S}$ *of unital associative* k*-algebra structures* $A_s = (P_s, \mu_s, \eta_s)$*, one on each position space. The bijection is* $\delta_{\mathrm{shape}} = \Delta$ *(the diagonal),* $\delta_s^\sharp = \mu_s$*,* $\varepsilon_s^\sharp = \eta_s$*. On morphisms this upgrades to an isomorphism of categories*

$$\mathrm{Comon}_{\lhd}\big(\mathrm{Fam}(\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}})\big) \;\cong\; \mathrm{Fam}(\mathbf{Alg}_k^{\mathrm{op}}),$$

the cocommutative comonoids forming the full subcategory $\mathrm{Fam}(\mathbf{CAlg}_k^{\mathrm{op}})$*.*

Proved in `proofs/2026-08-19-vec-comonoids-algebras.md` (registry node `part3-comonoid-algebras`, grade **proved**; hypotheses sharpened to finite-dimensional positions only, S arbitrary). The counit forces the shape comultiplication to the diagonal (the unique $(\mathbf{Set}, \times)$-comonoid on S); contravariance turns $\delta_s^\sharp$ into a multiplication $P_s \otimes P_s \to P_s$ and $\varepsilon_s^\sharp$ into a unit $k \to P_s$; the comonoid laws become associativity and the two-sided unit law.

Remark 5 (the algebroid guess, refuted in finite dimension; **proved**). Over **Set** the analogous fact is $\lhd$-comonoid $\cong$ small category (Ahman–Uustalu; **Cont** $\cong$ Fam($\mathbf{Set}^{\mathrm{op}}$)). One might hope $\lhd$-comonoid over **Vec** $\cong$ k-linear category (algebroid, Mitchell). This is *false* in finite dimension: because the counit forces $\delta_{\text{shape}} = \Delta$, the comultiplication never reaches an off-diagonal block $P_a \otimes P_b$ $(a \neq b)$, so there is no hom-between-distinct-objects composition. The structure is a disjoint family of one-object k-linear categories = k-algebras. A genuine algebroid requires double-indexed positions and a matrix (bimodule) composition $(P \circ Q)_{a,c} = \bigoplus_b P_{a,b} \otimes Q_{b,c}$ — the $\bigoplus_b$ restoring the extensivity coproduct whose absence caused the strict collapse. That identification is classical (Bénabou; Mitchell) and is treated separately; the finite single-index result stands as above.

Which monoidal / closed structures survive (high level)

Three companion results from the wider "containers over a base" programme locate the collapse inside the closed-structure theory. They are stated here at high level and *not* re-proved; each is proved in its own source file.

- **T1 (unit connectedness; proved).** For a closed symmetric monoidal cocomplete base $\mathcal{C}$, the extension functor $[\![-]\!]\colon \mathrm{Fam}(\mathcal{C}^{\mathrm{op}}) \to [\mathcal{C}, \mathcal{C}]$ is full and faithful *iff* the monoidal unit is *connected* ($\mathcal{C}(I, -)$ preserves coproducts). Over **Vec** the unit is *disconnected* (the zero object makes $|-|\colon \mathbf{Vec}_{\mathrm{fd}} \to \mathbf{Set}$ fail to preserve coproducts), which is exactly why $[\![-]\!]$ is not full — a base-general explanation of §2.
- **T2 (Dirichlet / Day tensor; proved).** The pointwise Dirichlet tensor on Fam($\mathcal{C}^{\mathrm{op}}$) is Day convolution, and its closedness is controlled by a familial-representability condition that fails over both **Vec** and $\mathbf{Vec}_{\mathrm{fd}}$.
- **T4-left (left closure; proved).** The composition product $\lhd$ is left-closed on the copower-tiny (collapse) locus, where $p \lhd q = p \otimes q$; this is the locus on which Proposition 2 lives.

(These are internal results of MacBeth's programme; grades are as recorded in memory and are cited, not re-derived, here.)

5 Composition and admissibility

Proposition 2 defines $\lhd$ on *finite-dimensional* positions. The general question is whether Fam($\mathbf{Vec}^{\mathrm{op}}$) carries a composition product at all.

Definition 3 ($\lhd$-admissibility)**.** Fam($\mathcal{C}^{\mathrm{op}}$) is $\lhd$-*admissible* if the image of $[\![-]\!]$ in $[\mathcal{C}, \mathcal{C}]$ is closed under composition: for all p, q there is r with $[\![r]\!] \cong [\![p]\!] \circ [\![q]\!]$.

Proposition 3 (admissibility is per-object; **proved**)**.** *Call* $P \in \mathcal{C}$ absorptive *if* $X \mapsto [P, [\![q]\!]X]$ *is an extension for every* q*. Then* Fam($\mathcal{C}^{\mathrm{op}}$) *is* $\lhd$*-admissible* ***iff*** *every object of* $\mathcal{C}$ *is absorptive. There are two sources of absorptivity:* copower-tiny *(flexible) objects, where* $[P, \coprod_t Y_t] = \coprod_t [P, Y_t]$*; and* distributive/extensive *(rigid) objects, via the* **Set***-style mechanism.*

Proposition 6.1 of `proofs/2026-09-01-gap1-setxvec-proved.md` (**proved**). In $\mathbf{Vec}_{\mathrm{fd}}$ every object is copower-tiny (finite-dimensional = finitely generated = copower-tiny; Lemma 5.1 there, **proved**), so Fam($\mathbf{Vec}_{\mathrm{fd}}^{\mathrm{op}}$) is admissible on the collapse locus with $\lhd = \otimes_k$ — this is Proposition 2 again.

5.1 A proved positive result: $\mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}}$ is admissible

Theorem 6 (Gap 1 inhabited; **proved**). *On the finite-vec-support locus $\mathcal{P} \subseteq \mathrm{Fam}((\mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}})^{\mathrm{op}})$ (shape S and set-positions A_s arbitrary; $\{s : V_s \neq 0\}$ finite), the base $\mathcal{C} = \mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}}$ is $\lhd$-admissible: for all $p, q \in \mathcal{P}$ there is $r = p \lhd q \in \mathcal{P}$ and an* explicit natural isomorphism $\Theta \colon [\![p \lhd q]\!] \Rightarrow [\![p]\!] \circ [\![q]\!]$ *— both components built as maps (the* **Set***-component a bijection with explicit inverse, the* **Vec***-component a natural composite of hom-out-of-coproduct, tensor–hom, and finite-support isomorphisms),* not *matched merely by cardinality.*

Theorem 3.1 of `proofs/2026-09-01-gap1-setxvec-proved.md` (**proved**; the map was built and its bijectivity, invertibility, and naturality confirmed computationally with zero failures). The base $\mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}}$ is non-collapse, non-cartesian, and has a disconnected unit, so it exhibits admissibility *without* unit-connectedness: it factors a rigid (extensive) **Set** part against a flexible (copower-tiny) **Vec** part. One honesty caveat, itself proved: the product $\lhd$ carries *no canonical* monoidal structure — the construction must choose a concentration slot — and the obstruction to canonicity is exactly the non-injectivity of $[\![-]\!]$ on objects (Theorem 2). A non-canonical monoidal structure does exist.

5.2 The open frontier: is full Vec admissible? (Question (Q), OPEN)

For *finite-dimensional* positions everything works. For *full* **Vec** the issue is that an infinite-dimensional position P is not copower-tiny, so $[P, \bigoplus_t(-)]$ need not be an extension. By Proposition 3 the whole question reduces to a single test object. The sharpest one is $E := k^{(\mathbb{N})}$ (countable-dimensional, free, non-tiny) against $q = (\mathbb{N}, (k))$, giving $[\![q]\!]X = \bigoplus_{\mathbb{N}} X$ and

Question 1 (the frontier; **OPEN**). Is

$$F(X) \;:=\; \mathrm{Hom}\big(E,\; \textstyle\bigoplus_{\mathbb{N}} X\big) \;=\; \textstyle\prod_{\mathbb{N}}\Big(\bigoplus_{\mathbb{N}} X\Big)$$

(row-finite $\mathbb{N} \times \mathbb{N}$ matrices over X)

naturally isomorphic to a coproduct of representables $\bigoplus_j \mathrm{Hom}(N_j, -)$ in the additive functor category $\mathrm{Add}(\mathbf{Vec}, \mathbf{Vec})$? Equivalently: is F a *projective* object of $\mathrm{Add}(\mathbf{Vec}, \mathbf{Vec})$?

The stakes: **(Q) YES** would give an irreducible admissible base beyond Theorem 6; **(Q) NO** would confirm the "absorptive dichotomy" conjecture (every nice admissible base decomposes as rigid-extensive $\times$ flexible-additive). *This is the one deep open problem in the theory and we state it as unsettled.*

What *is* proved is a precise homological reduction, all in ZFC, in `proofs/2026-09-02-vec-admissibility-rigidity.md` and `proofs/2026-09-04-Q-homological-reduction-and-target-dimension-dichotomy.md`:

- **Lemma R (Yoneda rigidity; proved).** Every $\alpha \in \mathrm{Nat}(F, \mathrm{id})$ has $\alpha \circ \mathrm{in}_n = 0$ for all but finitely many rows n. The engine is pure Yoneda: $\prod_{\mathbb{N}} = h_E$ and $\mathrm{Nat}(h_E, \mathrm{id}) = E = k^{(\mathbb{N})}$ is finite-support. This resurrects Baer–Specker-type rigidity *over a field*, where module slenderness fails.
- **Theorem E (homological reduction; proved, ZFC).** With $F_{\mathrm{fin}} := \bigoplus_n A \subseteq F$ ($A := \bigoplus_m \mathrm{id}$) and $Q := F/F_{\mathrm{fin}}$, for every M there is an exact sequence $0 \to \mathrm{coker}(\rho_M) \to \mathrm{Ext}^1(Q, M) \to \mathrm{Ext}^1(F, M) \to 0$. Hence

$$F \text{ is projective} \iff \mathrm{Ext}^1(F, M) = 0 \text{ for all } M.$$

  So (Q) is exactly the single homological question "is $\mathrm{Ext}^1(F, -) \equiv 0$?".

- **Theorem H (cardinality is neutral; proved, ZFC).** The id-dual $D(h_N) = \mathrm{Nat}(h_N, \mathrm{id}) = N$ has *no* $2^{\dim}$ blow-up (unlike $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}^{(\kappa)}, \mathbb{Z})$), so no cardinal invariant $\dim F(X)$ or $\dim \mathrm{Nat}(F, M)$ separates F from a free functor. Both classical decision routes — single-target rigidity (all such Nat-groups vanish on Q) and cardinality — are thereby *provably closed*; any resolution must engage $\mathrm{Ext}^1(F, -)$ at an infinite-dimensional target.

Remark 6 (status of (Q); **OPEN**, conjectured independent of ZFC). Question 1 is **not settled**. Because both the cardinality engine (Theorem H) and the single-target engine are provably unavailable over a field, the residual is a Whitehead-type extension problem on the syzygies of F. The current *conjecture* — theorem-supported but *not* proved — is that (Q) is **independent of ZFC**: plausibly NO under $2^{\aleph_0} < 2^{\aleph_1}$ and possibly YES under a failure of that hypothesis plus a uniformisation. Settling (Q) requires genuine set-theoretic input (Shelah; Eklof–Mekler transported to functor categories) and is the flagged frontier of the programme.

6 Formalisation

The object-level collapse (Theorem 2) is machine-checked in Lean 4 core, with *no* Mathlib dependency, in `lean/Containers/Containers/VecCollapse.lean`. Positions are carried by their finite basis types (k^2 by `Bool`, k by `Unit`), and the linear extension is modelled as the Π-over-shapes `LExt` $(S, \mathrm{Pos})\, X = (s : S) \to (\mathrm{Pos}\, s \to X)$ — the single $\Sigma \rightsquigarrow \Pi$ shift that *is* the biproduct collapse. The two containers $p = (\{*\}, k^2)$ and $p' = (\{1, 2\}, k)$ appear as `VecCollapse.p` and `VecCollapse.p'`. The theorem `Containers.VecCollapse.collapseIso` exhibits a full natural isomorphism `LinNatIso p p'` (both directions, both triangle identities, and naturality in X), while `Containers.VecCollapse.shapes_not_equiv` certifies (`sorry`-free, and axiom-free — `#print axioms` reports `[]`) that the shape sets `Unit` and `Bool` admit no bijection. Together they are the machine-checked witness that $[\![-]\!]$ presents the same functor $X \mapsto X \oplus X$ by two non-isomorphic containers — Theorem D(ii), **lean-verified**. The isomorphism `collapseIso` depends only on `Quot.sound` (function extensionality).

7 Summary and outlook

Statement	**Source**	**Grade**
Extension $[\![S,P]\!]W = \bigoplus_s \mathbf{Vec}(P_s, W)$; $\Sigma \to \Pi$ biproduct form	2026-08-18, Def. 0.2	definition
Finite collapse $[\![S,P]\!] \cong \mathrm{Id}^N$; classified by N alone (Thm 1)	2026-08-18, Thm 2.1	**proved**
Witness $(\{*\}, k^2) \ncong (\{1,2\}, k)$, same functor $X \mapsto X \oplus X$ (Thm 2)	2026-08-18 / VecCollapse.lean	**proved**, **lean-verified**
$[\![-]\!]$ neither full nor faithful; faithful iff $\lvert T\rvert \le 1$ or $S = \varnothing$ (Thm 3)	2026-08-18, Thm 3.3	**proved**
$\mathbb{F}_q$ defect $\delta = \prod_s a_s - \prod_s (a_s - (\lvert T\rvert - 1))$, $a_s = \sum_t q^{e_t d_s}$ (Thm 4)	2026-09-11, Thm 5	**proved**
Composition $(S,P) \lhd (T,Q) = (S \times T, (P_s \otimes Q_t))$ (Prop 2)	2026-08-18, Prop 4.1	**proved**
$\lhd$-comonoids $\cong$ families of k-algebras (Thm 5)	2026-08-19, Thm 7	**proved**
Admissibility $\Leftrightarrow$ every object absorptive (Prop 3)	2026-09-01, Prop 6.1	**proved**
$\mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}}$ is $\lhd$-admissible (Thm 6)	2026-09-01, Thm 3.1	**proved**
Homological reduction: F proj. $\Leftrightarrow \mathrm{Ext}^1(F,-) \equiv 0$ (Thm E)	2026-09-04, Thm E	**proved** (ZFC)
No cardinal invariant separates F from free (Thm H)	2026-09-04, Thm H	**proved** (ZFC)
(Q) Is full **Vec** admissible / F projective? (Question 1)	2026-09-02/04	**OPEN** (conj. indep. ZFC)

Outlook. The theory of vector-space containers is complete and clean on the finite-dimensional locus: the extension functor is neither full nor faithful (faithful only when the target has ≤ 1 shape or the source is empty) and is not injective on objects, the failure is exactly the non-extensivity of **Vec**, comonoids are families of algebras, and $\mathbf{Set} \times \mathbf{Vec}_{\mathrm{fd}}$ realises admissibility off the connected-unit pole. The single deep frontier is Question (Q) — whether *infinite-dimensional* linear resources can still absorb external shape — now sharpened to a precise, ZFC-reduced homological question ($\mathrm{Ext}^1(F,-) \equiv 0$?) with both classical decision routes provably closed and independence of ZFC conjectured. Settling it is the natural next step and would either produce a genuinely new irreducible composition base or complete the absorptive-dichotomy classification.

Draft — work in progress. Prepared for Neil Ghani / external requester, 2026-09-05. Internal sources: `proofs/2026-08-18-linear-containers-vec.md`, `proofs/2026-08-19-vec-comonoids-algebras.md`, `proofs/2026-09-11-collapse-defect-formula.md`, `proofs/2026-09-01-gap1-setxvec-proved.md`, `proofs/2026-09-02-vec-admissibility-rigidity.md`, `proofs/2026-09-04-Q-homological-reduction-and-target-dimension-dichotomy.md`, `lean/Containers/Containers/VecCollapse.lean`. *Grades per the corresponding registry files.*